

**A TECHNICAL REPORT ON STUDENTS INDUSTRIAL WORK
EXPERIENCE SCHEME (SIWES)**

**HELD AT
IMPERIAL MEMORIAL ACADEMY,
MUBI, ADAMAWA STATE**

**FROM: OCTOBER, 2023
TO: JANUARY, 2024**

**BY
MUSA A. ISA
(ST/CS/ND/22/366)**

**A SIWES REPORT SUBMITTED TO THE DEPARTMENT OF
COMPUTER SCIENCE, SCHOOL OF SCIENCE AND TECHNOLOGY
FEDERAL POLYTECHNIC, MUBI, ADAMAWA STATE, NIGERIA
IN PARTIAL FUFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF NATIONAL DIPLOMA (ND) IN COMPUTER SCIENCE**

MARCH, 2024

ATTESTATION

I **MUSA A. ISA** hereby attest that I did my Industrial Attachment Program at **Imperial Memorial Academy, Mubi, Adamawa State** between **October, 2023**, to **January, 2024**, and this report is prepared by me and submitted to the SIWES coordinator as required. I understand that all the information contained in this report are correct and will be penalized if found to be false. I also know that the SIWES coordinator may request for additional evidence as requested for reconciliation.

.....

MUSA A. ISA
(ST/CS/ND/22/366)

.....

DATE

CERTIFICATION

This is to certify that **MUSA A. ISA** with a registration number **ST/CS/ND/22/366** undergoes his industrial training **SIWES** at **Imperial Memorial Academy, Mubi, Adamawa State**. In partial fulfillment of the requirements for the award of National Diploma (**ND**) in computer science, Federal Polytechnic, Mubi, as certified by the undersign officer.

Mr. Hyellamada Simon

DEPARTMENTAL SIWES CO-ORDINATOR

.....

DATE

DEDICATION

I dedicate this technical report to my lovely parents who gave me all the support and care throughout my SIWES program.

ACKNOWLEDGMENTS

I thank God Almighty for making me to undergo students industrial work experience scheme (SIWES) successfully.

My gratitude goes to my parent for their prayers, financial and moral support during my attachment.

I also appreciate the kind gesture of my brothers and sisters and those who supports me in prayers and contribution during my industrial attachment.

I whole heartedly thank my Head of department Mallam Kassim Mustapha, SIWES coordinator Mr. Hyellamada Simon and all lectures of Computer Science for their effort to ensure my success as their students.

I am greatly indebted to my co-SIWES students to mention, may God strengthen our relationship together and grant us academic excellence.

I sincerely thank you all for your contribution and support.

TABLE OF CONTENTS

Title Page	-	-	-	-	-	-	-	-	-	i
Attestation	-	-	-	-	-	-	-	-	-	ii
Certification	-	-	-	-	-	-	-	-	-	iii
Dedication	-	-	-	-	-	-	-	-	-	iv
Acknowledgements	-	-	-	-	-	-	-	-	-	v
Table of contents	-	-	-	-	-	-	-	-	-	vi

CHAPTER ONE: INTRODUCTION

1.1	Brief History of SIWES	-	-	-	-	-	-	-	-	1
1.2	Aims and Objectives of SIWES	-	-	-	-	-	-	-	-	2
1.3	Roles of student	-	-	-	-	-	-	-	-	2
1.4	Objectives of the report	-	-	-	-	-	-	-	-	2

CHAPTER TWO: HISTORY OF THE ORGANIZATION

2.1	Organizational chart	-	-	-	-	-	-	-	-	3
2.2	Brief history of the organization	-	-	-	-	-	-	-	-	3

CHAPTER THREE: WORK EXPERIENCE I

3.1	Introduction to HTML	-	-	-	-	-	-	-	-	5
3.2	Introduction to CSS	-	-	-	-	-	-	-	-	8

CHAPTER FOUR: WORK EXPERIENCE II

4.1	Introduction to Networking	-	-	-	-	-	-	-	-	12
-----	----------------------------	---	---	---	---	---	---	---	---	----

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1	Summary	-	-	-	-	-	-	-	-	17
5.2	Conclusion	-	-	-	-	-	-	-	-	17
5.3	Recommendation	-	-	-	-	-	-	-	-	18
Reference	-	-	-	-	-	-	-	-	-	19

CHAPTER ONE: INTRODUCTION

1.1 BRIEF HISTORY OF SIWES

SIWES was founded in 1973 by ITF (Industrial Training Funds) to address the problem of tertiary institution graduates' lack of appropriate skills for employment in Nigerian industries. The Students' Industrial Work Experience Scheme (SIWES) was founded to be a skill training programme to help expose and prepare students of universities, Polytechnics and colleges of education for the industrial work situation to be met after graduation.

This system facilitates the transfer from the classroom to the workplace and aids in the application of knowledge. The program allows students to become acquainted with and exposed to the experience required in handling and operating equipment and machinery that are typically not available at their schools.

Prior to the establishment of this scheme, there was a rising concern and trend among industrialists that graduates from higher education institutions lacked appropriate practical experience for employment. Students who entered Nigerian universities to study science and technology were not previously trained in the practical aspects of their chosen fields. As a result of their lack of work experience, they had difficulty finding work.

As a result, employers believed that theoretical education in higher education was unresponsive to the needs of labor employers. Thousands of Nigerians faced this difficulty till 1973. The fund's main motivation for establishing and designing the scheme in 1973/74 was launched against this context.

The ITF (Industrial Training Fund) organization decided to aid all interested Nigerian students and created the SIWES program. The federal government officially approved and presented it in 1974. During its early years, the scheme was entirely supported by the ITF, but as the financial commitment became too much for the fund, it withdrew in 1978. The National Universities Commission (NUC) and the National Board for Technical Education (NBTE) were given control of the scheme by the federal government in 1979. The federal government handed over supervision and implementation of the scheme to ITF in November 1984. It was taken over by the Industrial Training Fund (ITF) in July 1985, with the federal government bearing entire responsibility for funding.

1.2 AIMS AND OBJECTIVES OF SIWES

The Industrial Training Fund's Policy Document No. 1 of 1973 which established SIWES outlined the objectives of the scheme as:

- i. Provide an avenue for students in Institutions of higher learning to acquire industrial skills and experience in their respective courses of study.
- ii. Prepare students for the Industrial Work situation they are likely to experience after graduation.
- iii. Expose students to work methods and techniques of handling equipment and machinery that may not be available in their Institutions.
- iv. Make the transition from school to the world of work easier; and enhance students' networks for later job placements.
- v. Provide students with an opportunity to apply their knowledge to real work situations, thereby bridging the gap between theory and practice; and vi. Enlist and strengthen Employers' involvement in the entire educational process; thereby preparing the students for employment in Industry and Commerce.

1.3 ROLES OF STUDENT

- i. Attend SIWES orientation programme before going on attachment.
- ii. Comply with the establishment's rule and regulation.
- iii. Arrange living accommodation during the period of attachment.
- iv. Record all training activity done and other assignment in the log book.
- v. Complete SPEI from ITF, FORM 8 and get it endorsed by the employer for submission to the ITF.

1.4 OBJECTIVES OF THE REPORT

The objectives of the SIWES report are;

- i. To make thorough explanation of the work done during my four-month industrial training.
- ii. To fulfill the requirement for national diploma in computer science.
- iii. To contribute to the body of knowledge and to enhance the understanding of the writer about a similar or same job.

CHAPTER TWO: HISTORY OF THE ORGANIZATION

2.1 ORGANIZATIONAL STRUCTURE

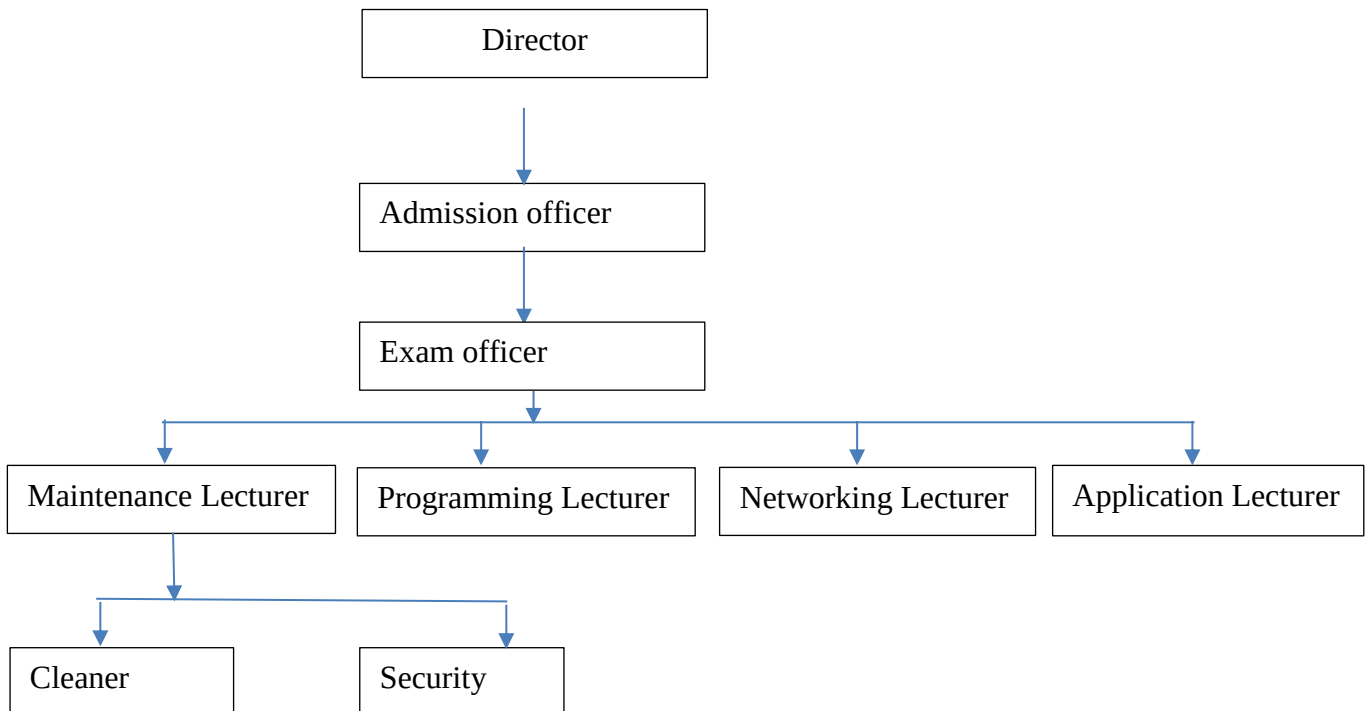


Figure 2.1: Organizational Chart

2.2 BRIEF HISTORY OF AMJAD INTEGRATED SERVICES

AMJAD is a leading web development company. It provides all kinds of mobile application development, web application development, cloud server administration services to its clients. Our main focus is to build awesome products for our clients to maintain their business more secured, fast, maintainable and reliable.

HISTORY

AMJAD INTEGRATED SERVICES NIG LTD is registered with Federal Republic of Nigeria under Corporate Affairs Commission by the government. Our registration number is BN248590. This company has 50 desktop Computers and 3 laptops with 7 numbers of staff, this company is located at Lokuwa near Emir's Palace GSS junction.

CHAPTER THREE: WORK EXPERIENCE I

3.1 INTRODUCTION TO HTML

What is HTML?

- i. HTML is the standard markup language for creating Web pages.
- ii. HTML stands for Hyper Text Markup Language
- iii. HTML describes the structure of Web pages using markup
- iv. HTML elements are the building blocks of HTML pages
- v. HTML elements are represented by tags
- vi. HTML tags label pieces of content such as "heading", "paragraph", "table", and so on.
- vii. Browsers do not display the HTML tags, but use them to render the content of the page

HTML Tags

HTML tags are element names surrounded by angle brackets:

`<tagname>content goes here...</tagname>`

- i. HTML tags normally come in pairs like `<p>` and `</p>`
- ii. The first tag in a pair is the start tag, the second tag is the end tag
- iii. The end tag is written like the start tag, but with a forward slash inserted before the tag name

WEB BROWSERS

The purpose of a web browser (Chrome, IE, Firefox, Safari) is to read HTML documents and display them.

The browser does not display the HTML tags, but uses them to determine how to display the document:

HTML VERSIONS

Since the early days of the web, there have been many versions of HTML:

Version	Year
HTML	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML5	2014

HTML EDITORS

Web pages can be created and modified by using professional HTML editors.

- i. Notepad (Windows) or TextEdit (Mac).
- ii. Sublime
- iii. Notepad ++

iv. Atom, etc.

We believe using a simple text editor is a good way to learn HTML.

Follow the four steps below to create your first web page with Notepad or TextEdit.

HTML PARAGRAPHS

HTML paragraphs are defined with the <p> tag:

Example

<p>This is a paragraph.</p>

<p>This is another paragraph.</p>

HTML HEADINGS

Headings are important in HTML documents.

Headings are defined with the <h1> to <h6> tags.

<h1> defines the most important heading. <h6> defines the least important heading.

Example

<h1>This is heading 1</h1>

<h2>This is heading 2</h2>

<h3>This is heading 3</h3>

<h4>This is heading 4</h4>

<h5>This is heading 5</h5>

<h6>This is heading 6</h6>

HTML Text Formatting Elements

Tag	Description
	Defines bold text
	Defines emphasized text
<i>	Defines italic text
<small>	Defines smaller text
	Defines important text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<ins>	Defines inserted text
	Defines deleted text
<mark>	Defines marked/highlighted text

HTML COMMENTS

Comment tags are used to insert comments in the HTML source code.

You can add comments to your HTML source by using the following syntax:

<!-- Write your comments here -->

With comments you can place notifications and reminders in your HTML:

Example

```
<!-- This is a comment -->  
<p>This is a paragraph.</p>  
<!-- Remember to add more information here -->  
»
```

HTML COLORS

In HTML, a color can be specified by using a color name, an RGB value, or a HEX value.

Color Names

In HTML, a color can be specified by using a color name:

Example

Color Name

Red
Orange
Green
Black

HTML supports 140 standard color names.

RGB Value

rgb(255,0,0)
rgb(255,255,0)
»

HEX Value

In HTML, a color can also be specified using a hexadecimal value in the form: #RRGGBB, where RR (red), GG (green) and BB (blue) are hexadecimal values between 00 and FF (same as decimal 0-255). For example, #FF0000 is displayed as red, because red is set to its highest value (FF) and the others are set to the lowest value (00).

HTML IMAGES

JPG Images

GIF Images

PNG Images

Example

```
<!DOCTYPE html>  
<html>  
<body>  
<h2>Spectacular Mountain</h2>  

```

</body>

</html>

HTML Images Syntax

In HTML, images are defined with the tag. The tag is empty, it contains attributes only, and does not have a closing tag. The src attribute specifies the URL (web address) of the image:

```

```

If a browser cannot find an image, it will display the value of the alt attribute:

Example

```

```

»

HTML LISTS

HTML List Example

An Unordered List:

- Item
- Item
- Item
- Item

An Ordered List:

1. First item
2. Second item
3. Third item
4. Fourth item

Unordered HTML List

An unordered list starts with the tag. Each list item starts with the tag.

The list items will be marked with bullets (small black circles) by default:

Example

```
<ul>
```

```
<li>Coffee</li>
```

```
<li>Tea</li>
```

```
<li>Milk</li>
```

```
</ul>
```

Ordered HTML List

An ordered list starts with the tag. Each list item starts with the tag.

The list items will be marked with numbers by default:

Example

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Ordered HTML List - The Type Attribute

The type attribute of the tag, defines the type of the list item marker:

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

3.2 INTRODUCTION TO CSS

- i. **CSS** stands for **Cascading Style Sheets**
- ii. CSS describes **how HTML elements are to be displayed on screen, paper, or in other media**
- iii. CSS **saves a lot of work**. It can control the layout of multiple web pages all at once
- iv. External stylesheets are stored in **CSS files**
- v. CSS is a language that describes the style of an HTML document.

CSS Example

```
body {
  background-color: lightblue;}

h1 {
  color: white;
  text-align: center;}

p {
  font-family: verdana;
  font-size: 20px;}
```

Note: that CSS code is simply written instructions that tells Web browsers (like FireFox and Internet Explorer) how to display things on a page. For example:

- i. make text bold.
- ii. position things a page.
- iii. set the font style for a page or paragraph etc.

3.2.1 APPLYING CSS

There are three ways to apply CSS to HTML: Inline, internal, and external.

Inline

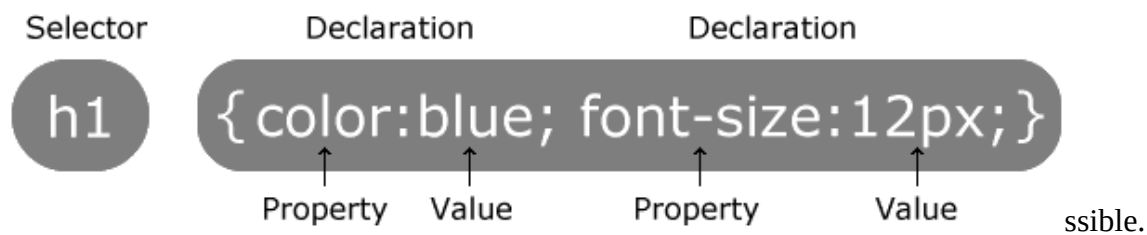
Inline styles are plonked straight into the HTML tags using the style attribute.

They look something like this:

```
<p style="color: red">text</p>
```

This will make that specific paragraph red.

But, if you remember, the best-practice approach is that the HTML should be a stand-alone, presentation free document, and so in-line styles should be avoided wherever possible.



Internal

Embedded, or internal, styles are used for the whole page. Inside the head element, the style tags surround all of the styles for the page.

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Example</title>
<style>
  p {
    color: red;
  }
  a {
    color: blue;
  }
</style>
```

This will make all of the paragraphs in the page red and all of the links blue.

Although preferable to soiling our HTML with inline presentation, it is similarly usually preferable to keep the HTML and the CSS files separate, and so we are left with our saviour.

External

External styles are used for the whole, multiple-page website. There is a separate CSS file, which will simply look something like:

```
p {
```

```
    color: red;
}
a {
    color: blue;
}
```

If this file is saved as “style.css” in the same directory as your HTML page then it can be linked to in the HTML like this:

```
<!DOCTYPE html>
<html>
<head>
    <title>CSS Example</title>
    <link rel="stylesheet" href="style.css">
```

3.2.2 CSS SYNTAX AND SELECTORS

CSS Syntax

A CSS rule-set consists of a selector and a declaration block:

The selector points to the HTML element you want to style. The declaration block contains one or more declarations separated by semicolons. Each declaration includes a CSS property name and a value, separated by a colon.

A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

Lengths and Percentages

There are many property-specific units for values used in CSS, but there are some general units that are used by a number of properties and it is worth familiarizing yourself with these before continuing.

- i. **px** (such as font-size: 12px) is the unit for pixels.
- ii. **em** (such as font-size: 2em) is the unit for the calculated size of a font. So “2em”, for example, is two times the current font size.
- iii. **pt** (such as font-size: 12pt) is the unit for points, for measurements typically in printed media.
- iv. **%** (such as width: 80%) is the unit for... wait for it... percentages.

CSS Selectors

CSS selectors are used to "find" (or select) HTML elements based on their element name, id, class, attribute, and more.

The Element Selector

The element selector selects elements based on the element name. You can select all <p> elements on a page like this (in this case, all <p> elements will be center-aligned, with a red text color):

Example

```
p {  
    text-align: center;  
    color: red;  
}
```

The id Selector

The id selector uses the id attribute of an HTML element to select a specific element. The id of an element should be unique within a page, so the id selector is used to select one unique element! To select an element with a specific id, write a hash (#) character, followed by the id of the element.

The style rule below will be applied to the HTML element with id="para1":

Example

```
#para1 {  
    text-align: center;  
    color: red;  
}
```

The Class Selector

The class selector selects elements with a specific class attribute. To select elements with a specific class, write a period (.) character, followed by the name of the class.

In the example below, all HTML elements with class="center" will be red and center-aligned:

Example

```
.center {  
    text-align: center;  
    color: red;  
}
```

CSS Comments

Comments are used to explain the code, and may help when you edit the source code at a later date.

A CSS comment starts with /* and ends with */. Comments can also span multiple lines:

Example

```
p {  
    color: red;  
    /* This is a single-line comment */  
    text-align: center;  
}
```

CHAPTER FOUR: WORK EXPERIENCE II

4.1 INTRODUCTION TO NETWORKING

Networking, also known as computer networking, is the practice of transporting and exchanging data between nodes over a shared medium in an information system. Networking comprises not only the design, construction and use of a network, but also the management, maintenance and operation of the network infrastructure, software and policies.

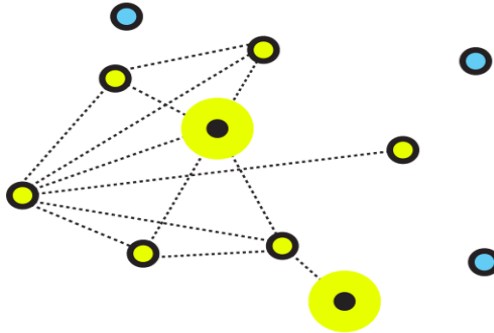


Figure 3.4: An illustrative image for networking

4.1.1 Types of Networking

There are two primary types of computer networking: wired networking and wireless networking. Wired networking requires the use of a physical medium for transport between nodes. Copper-based Ethernet cabling, popular due to its low cost and durability, is commonly used for digital communications in businesses and homes. Alternatively, optical fiber is used to transport data over greater distances and at faster speeds, but it has several tradeoffs, including higher costs and more fragile components.

4.1.2 Components of Networking

Computer networking requires the use of physical network infrastructure -- including switches, routers and wireless access points -- and the underlying firmware that operates such equipment. Other components include the software necessary to monitor, manage and secure the network.

4.1.3 Networking wiring colour code and crimping.

Purchasing Ethernet cables can be quite expensive and pre-made lengths are not always the length you need. Making Ethernet cables is easy with a box of bulk Category 5e Ethernet cable and RJ-45 connectors that are attached to the cut ends of your preferred cable length.

Bulk Ethernet Cable - Category 5e or CAT5e



Figure 3.5: Bulk Ethernet Cable - Category 5e or CAT5e

(You may also use Category 6 or CAT6 cabling which has higher performance specifications and is about 20% more expensive than CAT5e.)



Figure 3.6: Bulk RJ45 Crimpable Connectors for CAT-5e



Figure 3.7: RJ-45 Crimping tool

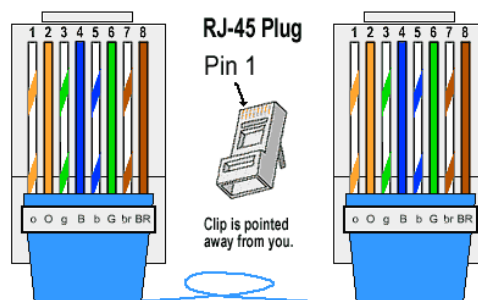


Figure 3.8: RJ45 diagram

4.1.4 Types of ethernet cables

There are two kinds of Ethernet cables you can make, **Straight Through** and **Crossover**. **Straight through** Ethernet cables are the standard cable used for almost all purposes, and are often called "patch cables". It is highly recommend you duplicate the color order as shown on the left. Note

how the green pair is not side-by-side as are all the other pairs. This configuration allows for longer wire runs.

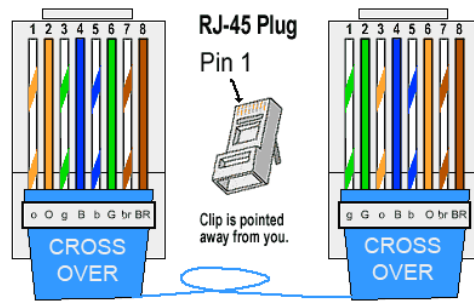


Figure 3.9: Straight through cable

Crossover Cables - The purpose of a Crossover Ethernet cable is to directly connect one computer to another computer (or device) without going through a router, switch or hub.

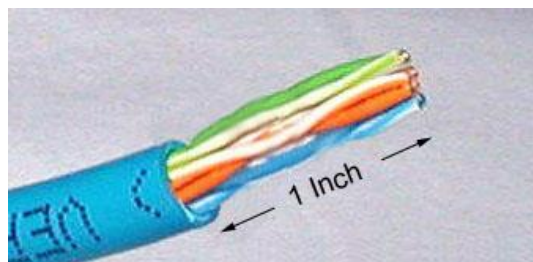


Figure 3.10: Cross over cable

Cut into the plastic sheath about 1 inch (2.5 cm) from the end of the cut cable. The crimping tool has a razor blade that will do the trick with practice.

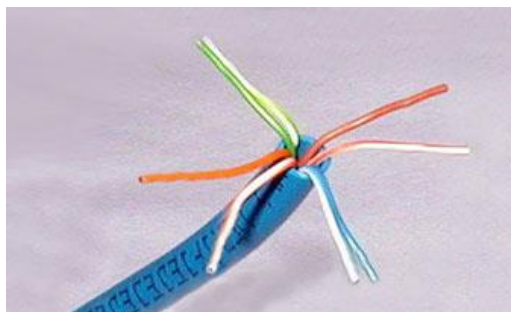


Figure 3.11: Unwind and pair the similar colors.

Pinch the wires between your fingers and straighten them out as shown. The color order is important to get correct.

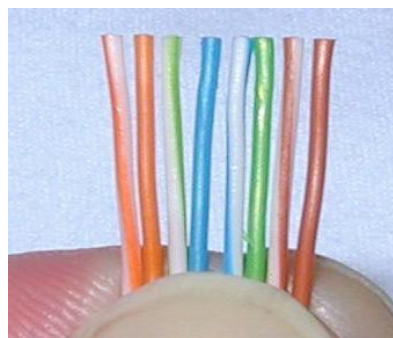


Figure 3.12: Colour order of cables

Use scissors to make a straight cut across the 8 wires to shorten them to 1/2 Inch (1.3 cm) from the cut sleeve to the end of the wires.



Figure 3.12: Scissors

Carefully push all 8 unstripped colored wires into the connector. Note the position of the blue plastic sleeve. Also note how the wires go all the way to the end.



Figure 3.13: Inserting the wires

A view from the top. All the wires are all the way in. There are no short wires.

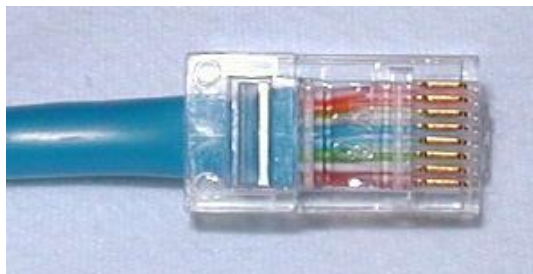


Figure 3.14: Top view of the inserted wires

Wrong Way - Note how the blue plastic sleeve is not inside the connector where it can be locked into place. The wires are too long. The wires should extend only 1/2 inch from the blue cut sleeve. Note how the wires do not go all the way to the end of the connector.

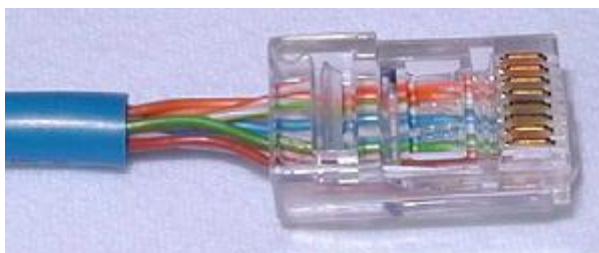


Figure 3.15: Wrong way of inserting the wires

Crimping the Cable ... carefully place the connector into the Ethernet Crimper and cinch down on the handles tightly. The copper splicing tabs on the connector will pierce into each of the eight wires. There is also a locking tab that holds the blue plastic sleeve in place for a tight compression fit. When you remove the cable from the crimper, that end is ready to use.



Figure 3.16: Crimping the cable

For a standard "Straight Through" cable, repeat all steps and wire color order on the other end of cable. For a cross-over cable, the other end will have a different color order as shown by the crossover picture above. Make sure to test the cables before installing them. An inexpensive Ethernet cable tester does this quite well.

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

The relevance of the SIWES program cannot be over emphasized considering the fact that it has significantly reduced the gap between my theoretical and practical knowledge about computer hardware and software, installations, maintenance and networking. The processes of communication which include data and telecommunication the use of switch in the networking and what networking is all about.

The program is indeed a commendable one in that it affords students ample opportunities of being exposed to good working relationship with colleagues and the field experience with customers. This little exposure has widened my knowledge about my course of study, not only that it has automatically changed my views about life in general. The firm at large has taught me how to be independent of my own how to be conscious of my health and safety at its peak relating to the environment where I carried out my SIWES program. It was indeed a highly rewarding experience to be with Amjad Integrated Service, Mubi, Adamawa State.

5.2 CONCLUSION

In conclusion, I thank ITF in general for their effort towards the Student Industrial Training Scheme. The contribution that the industrial training offered to student will not be over emphasized. It has exposed me seriously to a certain depth and length of practical capability on Web Designs.

It has also acquainted me with the working condition, which I am expected to encounter in the near future. I will say that SIWES has a greater advantage on me, it has greatly exposed me to the practical application of all that I have been through in the school, SIWES is an experience that all student must pass through this is because it gives a full practical knowledge of what has been through in classroom.

Finally, I have a strong believe that this comprehensive based on the experience, I acquired during the industrial training scheme will convince every user training is not difficult. I therefore strong conclude that the continuous existence of SIWES programme as it is very necessary since it plays a dominant role in the development of student of Computer Science in the acquisition of practical experience.

5.3 RECOMMENDATIONS

Below are the recommendations that should be given serious consideration so as alleviate the suffering of students undergoing SIWES.

Firstly, the Federal Government of Nigerian should make a positive effort in reducing the overall cost of production so that companies should be producing to fill capacity and accommodate SIWES populaces. Also, certain monthly allowance may be given to the student by company accepted then (student to ease transportation problem).

Secondly, the Industrial Training Fund (ITF), should try and increase the money paid at the end of SIWES to the student so as to justify the Cost of Living we experienced.

Thirdly, the period set for the SIWES should be increase to six mouths so as to enable the student involved and gain enough experience since it is widely believed that experience is the best teacher. Again, this period should also be in line with Nigerian University Commission curriculum for the university undergraduate student on industrial attachment.

Finally, the ITF official should please continue visiting the students, to ensure that what they are learning is in line with the ITF requirement.

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